Recitation 13

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May 26/27/28, 2021

Outline

1. **Review**

2. **Recitation**

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Review

- Central limit theorem

$$Z_n = \frac{S_n - \mathbf{E}[S_n]}{\sigma_{S_n}} = \frac{S_n - n\mathbf{E}[X]}{\sqrt{n}\sigma}$$

$$\mathbf{P}(Z_n \leq c) \to \Phi(c), \qquad \text{for all } c$$

- De Moivre-Laplace CLT
  - 1/2 correction for estimating discrete RVs

- The strong law of large numbers

$$\mathbf{P}\left(\lim_{n\to\infty} \frac{X_1 + \cdots + X_n}{n} = \mu\right) = 1$$

Outline

1. Review

2. Recitation

1. Consider the estimation of the probability p that a bulb produced by the factory is defectless. You are told to assume that all light bulbs have the same probability of having a defect, and that defects in different light bulbs are independent.

(a) Suppose that you test n randomly picked bulbs, what is a good estimate Z_n for p, such that Z_n converges to p in probability?

(b) If you test 50 light bulbs, what is the probability that your estimate is in the range $p \pm 0.1$?

(c) The management asks that your estimate falls in the range $p \pm 0.1$ with probability 0.95. How many light bulbs do you need to test to meet this specification?

Solution:

(a) Let X_i denote the test result of the i-th light bulb with PMF

$$p_{X_i}(x) = \begin{cases} p, & x = 1 \\ 1-p, & x = 0. \end{cases}$$

Then we have $\mathbf{E}[X_i] = p$ and $\mathrm{var}(X_i) = p(1-p)$. We further define Z_n as

$$Z_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

By using Chebyshev inequality,

$$\mathbf{P}(|Z_n - p| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

As $n \to \infty$, Z_n converges to p in probability.

(b) With Chebyshev inequality, we have

$$\mathbf{P}(|Z_{50} - p| \geq 0.1) \leq \frac{\sigma^2}{(0.1)^2 \cdot 50}.$$

Since X_i is a Bernoulli variable, $\sigma \leq \frac{1}{2}$. Thus,

$$\mathbf{P}(|Z_{50} - p| \geq 0.1) \leq 0.5 \implies \mathbf{P}(|Z_{50} - p| \leq 0.1) \geq 0.5.$$

(c) By using Chebyshev inequality,

$$\mathbf{P}(|Z_n - p| \geq 0.1) \leq \frac{\sigma^2}{(0.1)^2 n} \leq \frac{1}{0.04n} \leq 0.05,$$

which yields $n \geq 500$ to guarantee the estimation accuracy.

2. Random variable X is uniformly distributed between -1.0 and 1.0. Let $X_1, X_2, \ldots,$ be independent identically distributed random variables with the same distribution as X. Determine which, if any, of the following sequences (all with $i = 1, 2, \ldots$) are convergent in probability.

(a) X_i

(b) $Y_i = \frac{X_i}{i}$

(c) $Z_i = (X_i)^i$

Solution:

(a) No. X_i is uniformly distributed between -1.0 and 1.0.

(b) By using Chebyshev inequality, for $\epsilon > 0$,

$$\mathbf{P}(|Y_i - 0| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} = \frac{1/3}{i^2\epsilon^2}$$

As $i \to \infty$, Y_i converges to 0.

(c) By using Chebyshev inequality, for $\epsilon > 0$,

$$\lim_{i\to\infty} \mathbf{P}(|Z_i - 0| \geq \epsilon) = \lim_{i\to\infty} \mathbf{P}(X_i \geq \epsilon^{\frac{1}{i}}) + \lim_{i\to\infty} \mathbf{P}(X_i \leq -\epsilon^{\frac{1}{i}})$$
$$= \lim_{i\to\infty} (1 - \epsilon^{\frac{1}{i}}) \to 0$$

As $i \to \infty$, Z_i converges to 0.

3. A sequence X_n of RVs is said to converge to a number c **in the mean square**, if

$$\lim_{n\to\infty} \mathbf{E}[(X_n - c)^2] = 0.$$

(a) Show that convergence in the mean square implies convergence in probability.

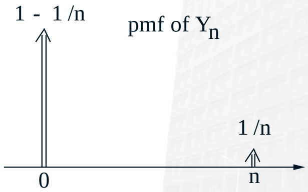
(b) Give an example to show that convergence in probability does not imply convergence in the mean square.

(a) With Markov inequality, we have

$$\mathbf{P}(|X_n - c| \geq \epsilon) = \mathbf{P}(|X_n - c|^2 \geq \epsilon^2) \leq \frac{\mathbf{E}[(X_n - c)^2]}{\epsilon^2}$$

Let $n \to \infty$. Because of convergence in the mean square, $\mathbf{E}[(X_n - c)^2] \to 0$. Therefore, $\mathbf{P}(|X_n - c| \geq \epsilon) \to 0$.

(b) See the example in Lecture 12 (P23).
Y_n converges in probability. However, $\mathbf{E}[Y_n^2] = n$, which diverges in the mean square.



4. We roll 10 fair dice. Please find the approximate probability that the sum obtained is between 30 and 45, inclusive.

4. We roll 10 fair dice. Please find the approximate probability that the sum obtained is between 30 and 45, inclusive.

Let X_i denote the value of the ith die, $i = 1, ..., 10$. Since X_i follows discrete uniform distribution between 1 and 6, we have

$$\mathbf{E}[X_i] = \frac{7}{2}, \quad \mathrm{var}(X_i) = \frac{35}{12}$$

Let $X = X_1 + \cdots + X_{10}$. With CLT and 1/2 correction, we have

$$\begin{aligned}
\mathbf{P}(30 \le X \le 45) &= \mathbf{P}(29.5 \le X \le 45.5) \\
&= \mathbf{P}\left(\frac{29.5 - 35}{\sqrt{10 \cdot \frac{35}{12}}} \le \frac{X - 35}{\sqrt{10 \cdot \frac{35}{12}}} \le \frac{45.5 - 35}{\sqrt{10 \cdot \frac{35}{12}}}\right) \\
&\approx \mathbf{P}(-1.0184 \le Z \le 1.9442) \\
&= \Phi(1.9442) + \Phi(1.0184) - 1
\end{aligned}$$